NU MOA

School of Information Technology
System Integration & Architecture Project

Project Overview

Each group will develop a **Full-Stack MERN Web Application** based on a real-world problem or business process. The project should demonstrate the group's ability to design, develop, test, and present a functional web-based system using **MongoDB, Express.js, React.js, and Node.js**.

The system should have a clearly defined **target client**, user roles, core business processes, and sufficient sample data to demonstrate its functionality. Each project must include the required system modules and successfully demonstrate the complete workflow during the final presentation.

The following sections outline the minimum requirements that must be included in the project:

Required System Modules:

Each project must include the following:

1. **Account Management**
   - User registration and login
   - Logout functionality
   - Profile/account information
   - Password Recovery

2. **User Management**
   - Admin can view registered users
   - User information can be added, updated, or managed
   - User roles and access permissions should be implemented where applicable

3. **Core Business Process Module**
   - The system must include main business process modules that represents the primary operation of the proposed project.
   - The module must support the **complete business workflow**, from creation/request through processing and completion.
   - The **core business process** should be appropriate to the proposed system. Examples include:
     1. Hotel Management – Reservation & Booking
     2. Restaurant Management – Order & Sales
     3. E-Commerce – Order Management
     4. Learning Management – Enrollment & Course Management
     5. Library System – Book Borrowing & Return
     6. Clinic System – Patient Appointment & Consultation
     7. Car Rental – Vehicle Rental & Reservation
     8. HR System – Employee Leave & Attendance

4. **Archive Module**
   - The system must have an **Archive** feature for storing records that are no longer active but still need to be retained.
   - Archived records should be separated from active records.
   - The system should provide a way to **view and, where appropriate, restore archived records**.
   - Students should demonstrate the archive functionality during the presentation.

5. **Required Users and Test Data**
   - The system must have at least **10 existing user accounts** before the presentation.
   - The accounts should contain realistic sample information and appropriate user roles.
   - The system must contain sufficient data to demonstrate the application's major features.

6. **Data Cleanup Before Presentation**
    - All unnecessary **test, dummy, and development data** must be removed before the final presentation.
    - Only the required demonstration data should remain in the system.
    - The 10 required users and 20 completed transactions must be **valid, realistic, and intentionally prepared for the demonstration**.

7. **Target Client**
    - Each group must identify a Target Client for their proposed system. The **target client** serves as a **sample organization or business** that the system is designed for. The client **does not need** to be an actual client or an organization that has agreed to use the system.

    - The target client should be relevant to the project's purpose and should help define the system's users, features, and business requirements.

    Example:

        - **Project:** Learning Management System
            - **Target Client:** Harvard University
        - **Project:** Restaurant Management System
            - **Target Client:** Jollibee
        - **Project:** Hotel Reservation System
            - **Target Client:** Marriott Hotel
        - **Project:** Agricultural Management System
            - **Target Client:** A local agricultural cooperative

Note: The Target Client is only a **sample/reference organization** for the project. Students are not required to obtain permission or establish an actual partnership with the organization.